

A Theory of Economic Slack

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APPENDIX D.

Geometry of the matching function

In this appendix, we establish several properties about the geometry of the matching function. Throughout we consider a matching function $m(s, b)$, with $s > 0$ and $b > 0$, that satisfies the following assumptions: the matching function is twice continuously differentiable, strictly concave in at least one argument (s or b), and with constant returns to scale.

D.1. Second derivatives of the matching function

In this section we establish geometric properties of the matching function based on the properties of its second derivatives.

D.1.1. Preliminary results

Since the matching function exhibits constant returns to scale, we can express it as

$$(D.1) \quad m(s, b) = s \cdot f(\theta),$$

where $f(\theta) = m(1, \theta)$ and $\theta = b/s$.

Using (D.1), we compute the first derivatives of the matching function. The partial derivative of the matching function with respect to s is

$$\frac{\partial m}{\partial s} = f(\theta) + s \cdot f'(\theta) \cdot \frac{\partial \theta}{\partial s}.$$

Since $\partial\theta/\partial s = -b/s^2 = -\theta/s$, the partial derivative simplifies to

$$(D.2) \quad \frac{\partial m}{\partial s} = f(\theta) - \theta f'(\theta).$$

Next, the partial derivative of the matching function with respect to b is

$$\frac{\partial m}{\partial b} = s \cdot f'(\theta) \cdot \frac{\partial \theta}{\partial b}.$$

As $\partial\theta/\partial b = 1/s$, the partial derivative simplifies to

$$(D.3) \quad \frac{\partial m}{\partial b} = f'(\theta).$$

From here, we compute the second derivatives of the matching function. Starting from (D.2), we obtain

$$\frac{\partial^2 m}{\partial s^2} = f'(\theta) \cdot \frac{\partial \theta}{\partial s} - \frac{\partial \theta}{\partial s} \cdot f'(\theta) - \theta \cdot f''(\theta) \cdot \frac{\partial \theta}{\partial s}.$$

Since $\partial\theta/\partial s = -\theta/s$, the derivative simplifies to

$$(D.4) \quad \frac{\partial^2 m}{\partial s^2} = \frac{\theta^2}{s} \cdot f''(\theta).$$

Then, starting from (D.3), we compute another second derivative:

$$\frac{\partial^2 m}{\partial b^2} = f''(\theta) \cdot \frac{\partial \theta}{\partial b}.$$

As $\partial\theta/\partial b = 1/s$, the derivative simplifies to

$$(D.5) \quad \frac{\partial^2 m}{\partial b^2} = \frac{1}{s} \cdot f''(\theta).$$

Starting from (D.3), we compute the cross-partial derivative of the matching function:

$$\frac{\partial^2 m}{\partial b \partial s} = f''(\theta) \cdot \frac{\partial \theta}{\partial s}.$$

Since $\partial\theta/\partial s = -\theta/s$, the cross-partial derivative becomes

$$(D.6) \quad \frac{\partial^2 m}{\partial b \partial s} = -\frac{\theta}{s} \cdot f''(\theta).$$

D.1.2. Main results

Using the preliminary results, we obtain two key properties of the matching function.

RESULT D.1. *The matching function is strictly concave in each argument: $\partial^2 m / \partial s^2 < 0$ and*

$$\partial^2 m / \partial b^2 < 0.$$

PROOF. The matching function is strictly concave in at least one argument, which means that either $\partial^2 m / \partial s^2 < 0$ or $\partial^2 m / \partial b^2 < 0$. However, equations (D.4) and (D.5) show us that either assumption is equivalent: namely, they both mean that $f''(\theta) < 0$. (Recall that $s > 0$ and $b > 0$ so $\theta^2/s > 0$ and $1/s > 0$.) Accordingly, if either one is true, the other is true too. We therefore have both $\partial^2 m / \partial s^2 < 0$ and $\partial^2 m / \partial b^2 < 0$. $\square$

RESULT D.2. *The matching function's two arguments are Edgeworth-Pareto complements: $\partial^2 m / \partial b \partial s > 0$. The cross-partial derivative is related to the own-partial derivatives by*

$$\frac{\partial^2 m}{\partial b \partial s} = \sqrt{\frac{\partial^2 m}{\partial b^2} \cdot \frac{\partial^2 m}{\partial s^2}}.$$

PROOF. Once again, the assumption that the matching function is strictly concave in at least one argument implies $f''(\theta) < 0$. With (D.6), this in turn implies $\partial^2 m / \partial b \partial s > 0$, which indicates Edgeworth-Pareto complementarity (Hicks 1946, p. 42).

Next, we establish the link between the cross-partial and own-partial derivatives. Equations (D.4) and (D.5) imply that

$$\frac{\partial^2 m}{\partial s^2} \cdot \frac{\partial^2 m}{\partial b^2} = \left[-\frac{\theta}{s} \cdot f''(\theta) \right]^2.$$

Since $-(\theta/s)f''(\theta) > 0$, the square root of the above equation is just $-(\theta/s)f''(\theta) > 0$, which from equation (D.6) we know is $\partial^2 m / \partial b \partial s$. $\square$

D.2. Hessian of the matching function

In this section we establish geometric properties of the matching function based on the properties of its Hessian. Using again equations (D.4), (D.5), and (D.6), we see that the Hessian is given by

$$H = \begin{bmatrix} \frac{\partial^2 m}{\partial s^2} & \frac{\partial^2 m}{\partial b \partial s} \\ \frac{\partial^2 m}{\partial b \partial s} & \frac{\partial^2 m}{\partial b^2} \end{bmatrix} = \frac{f''(\theta)}{s} \cdot \begin{bmatrix} \theta^2 & -\theta \\ -\theta & 1 \end{bmatrix}.$$

RESULT D.3. *The matching function is jointly concave in its two arguments, but not strictly concave.*

PROOF. To establish that the matching function is jointly concave, we show that its Hessian is negative semidefinite (result A.14). First, we see that the determinant of the Hessian is 0: $\det(H) \propto \theta^2 \times 1 - (-\theta) \times (-\theta) = \theta^2 - \theta^2 = 0$. So one eigenvalue is 0. The

trace of the Hessian is $\text{tr}(H) = (1 + \theta^2)f''(\theta)/s < 0$ since the assumption that the matching function is strictly concave in at least one argument implies $f''(\theta) < 0$. So the second eigenvalue is strictly negative (result A.5). Hence, the Hessian is negative semidefinite. However, because the Hessian is rank one, it is not negative definite, so the matching function is not strictly concave. $\square$

Bibliography

Hicks, John R. 1946. *Value and Capital: An Inquiry Into Some Fundamental Principles of Economic Theory*. 2nd ed. Oxford: Oxford University Press.